TIR = high denser – low denser

Blessed G/Michael Catholic Secondary School

2nd Semester Mathematics test two for Grade 10th

May, 2018 E.C Time allowed -1hr

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I. Choose the best answer among the given alternatives.

A ___ 1. A cone of 12m high is cute 6cm from the vertex to form a frustum of volume $63\pi cm^3$. Find the radius of the cone

A. $3\sqrt{2}$ cm B. $4\sqrt{2}$ cm C. $5\sqrt{2}$cm D. $6\sqrt{2}$ cm E none

B ___ 2. The lower base of a frustum of a regular pyramid is a square of side length 6cm. and the upper base has side length 4cm. If the slant height is 8cm then its total surface area equal to:------

A. $160cm^2$ B $212\ cm^2$ C. $222cm^2$ D. $52\ cm^2$ E. none

$144cm^3$ E ___ 3. A hemispherical solid has radius 6cm .what is its volume?

A. $196\pi\ cm^3$ B. $240\pi cm^3$ C. $288\pi cm^3$ D. $324\ \pi cm^3$ E. none

C ___ 4. The lateral edge of an equilateral triangular pyramid is 6cm .What is its lateral surface area?

A $6\sqrt{3}\ cm^2$ B. $9\sqrt{3}\ cm^2$ C. $27\sqrt{3}cm^2$ D. $18\ cm^2$ E. none

C ___ 5. The height of a right circular cylinder is equal to its base radius. If its volume is $27\pi cm^3$ then its radius is equal to _____

A. 1cm B. 2cm C.3cm D. 6cm E. none

I ---6. The lower base of a frustum of a right circular cone of height 6cm coincides with the upper base of a right circular cylinder of radius 8cm .If the height of the cylinder is is 10cm ,then the volume of the solid figure is

A. $260\ \pi cm^3$ B $340\pi cm^3$ C. $620\ \Pi cm^3$ D. $760\pi cm^3$ E. none

II. Short Answer(2 point each)

7. The lower base of a frustum of a regular pyramid is an equilateral triangle of side length 8cm and the upper base has aside length 4cm. If the slant height is 5cm then the total surface area of the frustum is _____

8. A spherical storage has radius 3cm,how many liter of oil will it hold? _____

9. The radius of the first conical vessel is twice the radius of the second one what is the ratio their volume if the ratio of the height of the 2nd to the 1st is 0.5 _____

10. A right circular cylinder of radius 6cm and altitude 8cm is placed on a frustum of a cone with lower base radius 9cm .If the height of the frustum is 4cm ,then,

a. What is the lateral area of the composed solid figure _____
b. What is the total surface area of the composed solid figure _____
c. What is the volume of this composed solid figure _____

6, $V_{cylinder} = Abh = \pi r^2 h = \pi(8^2)(10) = 64(10) = 640\pi cm^3$

$V_{frustum\ of\ cone} = \frac{1}{3}\pi h(r^2 + r'^2 + rr')$

$= \frac{1}{3}\pi(6)(8^2 + r'^2 + 8r')$

$= 2\pi(64 + 8r + r'^2)$

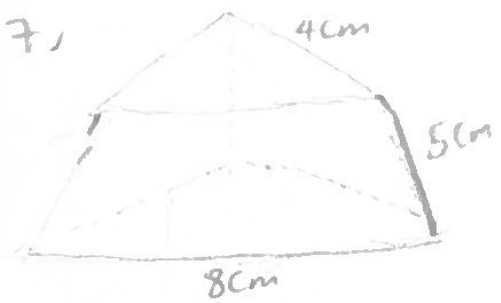
12
8
96

$V_{total} = 2\pi(64 + 8r + r'^2) + 640\pi cm^3$

$(128 + 16r + 2r'^2 + 640)\pi cm^3$

$(768 + 16r + 2r'^2)\pi cm^3$

7,



$LSA = \frac{1}{2}l(p' + p) = \frac{1}{2}5(12 + 24)$

$= \frac{5(36)}{2} = \frac{180}{2} = 90cm^2$

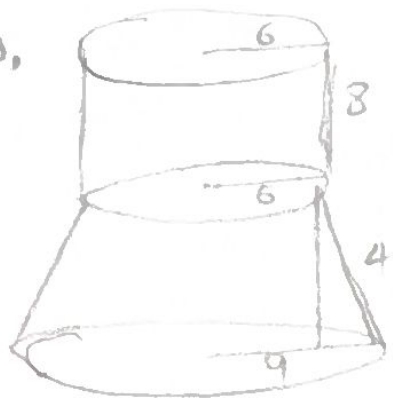
$B_{A\,upper} = \frac{1}{2}ab\sin 60^\circ = \frac{1}{2}4\cdot4\sin 60^\circ = \frac{1}{2}16\cdot\frac{\sqrt{3}}{2} = \frac{16\sqrt{3}}{4} = 4\sqrt{3}$

$B_{A\,lower} = \frac{1}{2}ab\sin 60^\circ = \frac{1}{2}8\cdot8\sin 60^\circ = \frac{1}{2}64\frac{\sqrt{3}}{2} = \frac{64\sqrt{3}}{4} = 16\sqrt{3}$

$BA = 4\sqrt{3} + 16\sqrt{3} = 20\sqrt{3}$

$TSA = LSA + BA = 90 + 20\sqrt{3}\ cm^2$

10,



$LSA_{cy} = 2\pi rh = 2(6)(8)\pi = 96\pi cm^2$

$LSA_{cone} = l\pi(r + r') = l = \sqrt{h^2 - (R-r)^2} = \sqrt{4^2 + (7-6)^2} = \sqrt{16+9} = 5$

$15(5)\pi = 75\ cm^2$

$LSA = 96 + 75 = 171\pi cm^2$

$TSA = BA_{cylinder} = \pi r^2 = \pi 36 cm^2$

$BA_{cone} = \pi r^2 = 81\pi cm^2$

$BA = 36 + 81 = 117\pi cm^2$

$LSA + BA = 171 + 117 = 288\pi cm^2$

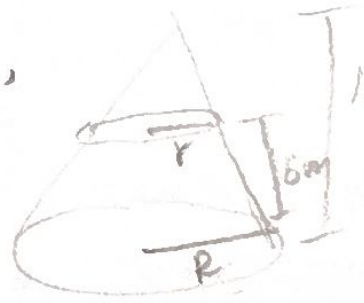
$V_{cylinder} = \pi r^2 h = 36(8)\pi = 288\pi cm^3$

$V_{cone} = \frac{1}{3}\pi h(r^2 + r'^2 + rr')$

$\frac{1}{3}\pi 4(6^2 + 9^2 + 6\cdot9) = 228\pi cm^3$

$V_{Total} = 288\pi + 228\pi = 516\pi cm^3$

1,



$\frac{r}{R} = \frac{6}{12} = \frac{r}{R} = \frac{1}{2} = R = 2r$

$V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$

$V = \frac{1}{3}\pi 6((2r)^2 + r^2 + r(2r))$

$3\cdot63\pi = \pi 6(4r^2 + r^2 + 2r^2)$

$\frac{189\pi}{\pi} = \frac{\pi 6(7r^2)}{\pi}$

$189 = 42\cdot r^2$

$r^2 = \sqrt{\frac{189}{42}} = \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$

$R = 2r = 2\cdot\frac{3\sqrt{2}}{2} = \frac{6\sqrt{2}}{2} = \underline{3\sqrt{2}}$